CLASS X MATHEMATICS — EXAM REVISION NOTES

8 chapters · formulas · worked solutions · graphs & constructions · every question of 3 papers solved

Built from the **BCSE 2025**, **2024** and **2023** Mathematics question papers (Final I). Every Section A item and every Section B part is solved inside.

How to use: Orange tags show where a topic appeared: P25 Q2a = 2025 paper question 2(a); P24 = 2024; P23 = 2023. Green boxes are **solved questions** with the mark value in brackets and the final answer in **red**. Each chapter starts with the **formulas and rules**, then the solved past-paper questions. Practise by covering the green box and solving first. Graphs and constructions are drawn to scale so you can copy the method.

⚠ **Read this first — limits of these notes** I had the three question papers only (no marking schemes), so answers are my own working, checked by calculation. Some questions are unclear or have printing faults and are marked **(check)**: **(a)** 2025 Q1a(xiv): the options give the ratio old : new (1 : 4) but the question asks new : old (4 : 1); **(b)** 2024 Q1a(xvii): the printed answer options contain only the *curved* surface of the cone (πrs), not the total surface; **(c)** 2025 Q6c: Option A costs only Nu 2,750 × 12 = Nu 33,000, less than the laptop price — probably a misprint; **(d)** 2023 Q1a(xv) ("order of turn symmetry") and 2023 Q2c, Q5b, Q9d depend on how the figure is read. In the exam, answer as printed and show full working — **method marks** are given even if the final number differs.

How marks are given (all three papers)

- **Section A (40 marks):** Q1 = **20 MCQs, 2 marks each**. Circle one letter. Many questions need a figure, a graph or a table, so practise reading them.
- **Section B (60 marks):** attempt **any SIX** of Q2–Q9 (10 marks each). Choose the questions you can finish: constructions, graphs and calculations are quickest to score.
- **Show every step:** formula → substitution → answer with **unit**. Marks are given for the method even if the last step is wrong.
- **Graphs:** ruled axes with scale, label, plotted points ×, smooth curve or ruled line, title. **Constructions:** keep all arcs and construction lines, label points, state the measurement.
- **"Justify / explain":** give a reason with a number from the question. **Formulae page:** the formula sheet is printed on page 33 of the paper — you do not need to memorise the volume and surface-area formulas, but you must know how to use them.

Paper pattern (2023, 2024, 2025)

Part	Content	Marks
Q1	20 multiple-choice questions (figures, graphs, tables)	40
Sec B Q2–Q9	8 questions of 10 marks: attempt any 6 (calculations, graphs, constructions, proofs)	60

Topic map: where each chapter appeared

Chapter	2023	2024	2025
1 Numbers, surds & finance	Q1ii,iii; Q2a; Q5a; Q7a; Q8a	Q1iii,xi; Q5c; Q6c; Q7c; Q9a	Q1v,vii; Q2b; Q4d,e; Q6c; Q9d
2 Linear equations & functions	Q1iv,vi,vii; Q2b; Q3b; Q6b	Q1iv,xiv; Q4c; Q6a; Q7b; Q9d	Q1iv,ix; Q2d; Q3c; Q6a; Q7c
3 Quadratics & transformations	Q1v,viii; Q4a; Q7b; Q8b	Q1x,xii,xviii; Q2c; Q3a; Q8c	Q1iii,xii,xv; Q5c,d; Q6d; Q9a
4 Matrices & networks	Q1i; Q4b; Q9a	Q1i,xv; Q3b	Q1i,vi; Q4c; Q9c
5 Trigonometry	Q1xii,xiii; Q4c; Q7d	Q1viii,xx; Q5d; Q8a	Q1viii,xi; Q2a; Q5b; Q8d
6 Measurement (solids)	Q1ix,x,xi,xiv; Q2c; Q5b; Q6a; Q8c; Q9b	Q1ii,xiii,xvii,xix; Q2b; Q4a; Q6b; Q8b; Q9c	Q1ii,xiv,xvii,xix; Q3b; Q6b; Q7b,d; Q8c
7 Geometry & symmetry	Q1xv,xvi; Q2d; Q3c; Q9c	Q1v,vii,ix; Q2a; Q4b	Q1x,xviii; Q3d; Q4b; Q7a; Q8a
8 Data & probability	Q1xvii–xx; Q3a; Q5c; Q6c; Q7c; Q8d; Q9d	Q1vi,xvi; Q3c; Q5a,b; Q7a; Q9b	Q1xiii,xvi,xx; Q2c; Q3a; Q4a; Q5a; Q8b; Q9b

Repeated every year: compound interest & dividend/yield • quadratic graphs, roots and transformations • matrix multiplication (points / cost tables) • trig ratios and proofs • volume & surface area, **efficiency** of solids • constructions (circumcircle, centroid) • turn symmetry • scatter plots and correlation • probability (with and without replacement, Venn).

1 · NUMBERS: SURDS & FINANCIAL MATHEMATICS

P25 Q1v,vii,Q2b,Q4d,e,Q6c,Q9d · P24 Q1iii,xi,Q5c,Q6c,Q7c,Q9a · P23 Q1ii,iii,Q2a,Q5a,Q7a,Q8a

Radicals (surds): rules

- **Product / quotient:** $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$; $\sqrt{a} \div \sqrt{b} = \sqrt{\frac{a}{b}}$. A radical is **simplified** when no square (or cube) factor is left under the root: $\sqrt{50} = 5\sqrt{2}$; $\sqrt[3]{40} = 2\sqrt[3]{5}$.
- **Like surds** add and subtract: $8\sqrt{y}$-type terms are first simplified, then the like surds are combined.
- **Entire radical:** put the number under the root: $a\sqrt{b} = \sqrt{a^2 b}$ ($3\sqrt{5} = \sqrt{45}$).
- **Powers of a root:** $\sqrt{x^n} = x^{n/2}$; $\sqrt{(2^m)}$ is a whole number when m is even and contains a radical when m is **odd**.
- **Perfect cube**: every prime exponent is a multiple of 3.

Interest & shares

Idea	Formula
Simple interest	$SI = \frac{P\,R\,T}{100}$; Amount = P + SI
Compound interest	$\mathbf{A = P(1 + \frac{r}{n})^{nt}}$ (r as a decimal; n = 1 yearly, 2 semi-annual, 4 quarterly, 12 monthly)
Dividend amount	**DA = fv × r × n** (fv face value, r dividend rate, n number of shares)
Yield %	$\mathbf{Yield = \frac{DA}{OI} \times 100\%}$ (OI = original investment = shares × price paid)
Premium / discount	$\text{premium \%} = \frac{(\text{market} - \text{face})}{\text{face}} \times 100$

- To **compare** two rates with different compounding, compare the **effective annual rate** $(1 + r/n)^n - 1$, or the amounts after the same time.

Solved questions

P25 Q1v [2] Perimeter = $2(l + w) = 12\sqrt{3} \rightarrow l + w = 6\sqrt{3}$. One side is $2\sqrt{12} = 4\sqrt{3}$. So the missing side = $6\sqrt{3} - 4\sqrt{3} = \mathbf{2\sqrt{3}}$ **(B)**.

P25 Q1vii [2] Face value = 200 × 50 = Nu 10,000. Rate = $\frac{3500}{10000} \times 100 =$ **35% (C)**.

P25 Q2b [3] Compound semi-annually: $35\,608 = 20\,000(1 + r/2)^8 \rightarrow (1 + r/2)^8 = 1.7804 \rightarrow 1 + r/2 = 1.0748 \rightarrow r/2 = 0.0748 \rightarrow r \approx 0.1495$. The interest rate was **about 15% per year** (semi-annually 7.5% per half-year).

P25 Q4d [2] Distance = $\sqrt{(7\sqrt{3})^2 + (4\sqrt{6})^2} = \sqrt{147 + 96} = \sqrt{243} =$ **$9\sqrt{3}$ km ≈ 15.6 km**. (This straight line is shorter than the road $7\sqrt{3} + 4\sqrt{6} \approx 21.9$ km, so it is the more fuel-efficient route.)

P25 Q4e [1] $\sqrt{2^m} = 2^{m/2}$. If m is **even** the exponent m/2 is a whole number, so no radical. If m is **odd**, e.g. $\sqrt{2^3} = 2\sqrt{2}$, a radical remains. So **the answer contains a radical when m is odd**.

P25 Q6c [2] Option B: $A = 60\,000(1 + 0.12/2)^2 = 60\,000 \times 1.1236 =$ **Nu 67,416**. Option A as printed: 12 × 2 750 = **Nu 33,000**. **(check)** 33 000 is less than the laptop price, so the figure is probably misprinted. Compare the totals: as printed, **Option A is cheaper (Nu 33,000 < Nu 67,416)**; if each instalment were large enough to cover the price, compare the two totals in the same way and choose the smaller.

P25 Q9d [2] $\sqrt{48} = 4\sqrt{3}$; $y = \sqrt{36x^m} = 6x^{m/2}$; so $x \times y = 24\sqrt{3}\,x^{m/2}$. $q = 24x^7\sqrt{3x} = 24\sqrt{3}\,x^{7.5}$. Equating the powers: $m/2 = 7.5 \rightarrow$ **m = 15**.

P24 Q1iii [2] $37\,960 = 30\,000(1 + r/2)^6 \rightarrow (1 + r/2)^6 = 1.2653 \rightarrow 1 + r/2 = 1.04 \rightarrow r = 0.08 =$ **8% (B)**.

P24 Q1xi [2] $\sqrt{8y} = 2\sqrt{2y}$; $5\sqrt{50y} = 25\sqrt{2y}$; $2\sqrt{18y} = 6\sqrt{2y}$. Sum = $(2 + 25 - 6)\sqrt{2y} =$ **$21\sqrt{2y}$ (B)**.

P24 Q5c [3] Entire radicals: $3\sqrt{5} = \sqrt{45}$; $5\sqrt{2} = \sqrt{50}$; $13 = \sqrt{169}$; $2\sqrt{3} = \sqrt{12}$. From least to greatest: **$\sqrt{12} < \sqrt{45} < \sqrt{50} < \sqrt{169}$**, i.e. **$2\sqrt{3}$, $3\sqrt{5}$, $5\sqrt{2}$, 13**.

P24 Q6c [3] (i) DA = 100 × 0.15 × 150 = **Nu 2,250**. (ii) Bought at 25% premium: price = 125 per share; OI = 150 × 125 = 18 750. Yield = $\frac{2250}{18750} \times 100 =$ **12%**. (iii) The savings account gives 7.25%. Since 12% > 7.25%, **no — the shares gave the better return**; he would not have benefited more with the savings account.

P24 Q7c [3] (i) $1 + \sqrt{2m + 3} = 6 \rightarrow \sqrt{2m + 3} = 5 \rightarrow 2m + 3 = 25 \rightarrow$ **m = 11**. (ii) $\sqrt{36x^a} \times \sqrt{48} = 6x^{a/2} \times 4\sqrt{3} = 24\sqrt{3}\,x^{a/2}$; RHS = $24\sqrt{3}\,x^{7.5} \rightarrow a/2 = 7.5 \rightarrow$ **a = 15**.

P24 Q9a [3] Effective rates: monthly 14.7%: $(1 + 0.147/12)^{12} - 1 =$ **15.73%**. Semi-annual 15%: $(1.075)^2 - 1 =$ **15.56%**. A borrower wants the **lower** cost, so choose **15% compounded semi-annually** (15.56% < 15.73%).

P23 Q1ii [2] $\sqrt[3]{40} = \sqrt[3]{(8 \times 5)} = 2\sqrt[3]{5} \rightarrow$ **D**.

P23 Q1iii [2] Interest on Nu 5000: A = 6% simple = 300; B = $5000(1.05) - 5000 = 250$; C = $5000(1.015)^2 - 5000 = 151.1$; D = $5000(1.00725)^4 - 5000 = 146.6$. Most is **Account A (A)**.

P23 Q2a [2] $K = 2^9 \times 3^6 \times 5^y$. 9 and 6 are multiples of 3, so K is a perfect cube when the exponent y is also a multiple of 3: **y = 0, 3, 6, 9, …** (any multiple of 3).

P23 Q5a [4] (i) Premium = $\frac{225 - 200}{200} \times 100 =$ **12.5%**. (ii) DA = 200 × 0.24 × 300 = **Nu 14,400**. (iii) OI = 300 × 225 = 67 500; yield = $\frac{14400}{67500} \times 100 =$ **21.33%**.

P23 Q7a [2] $\frac{\sqrt{5x^3} \times \sqrt{9x^4}}{\sqrt{80x}} = \sqrt{\frac{45x^7}{80x}} = \sqrt{\frac{9x^6}{16}} = \mathbf{\frac{3x^3}{4}}$.

P23 Q8a [2] Amount = 25 000 + 4 450 = 29 450 after 3 years, monthly: $25\,000(1 + r/12)^{36} = 29\,450 \rightarrow (1 + r/12)^{36} = 1.178 \rightarrow 1 + r/12 = 1.00456 \rightarrow r/12 = 0.00456 \rightarrow$ **r ≈ 5.5% per year**.

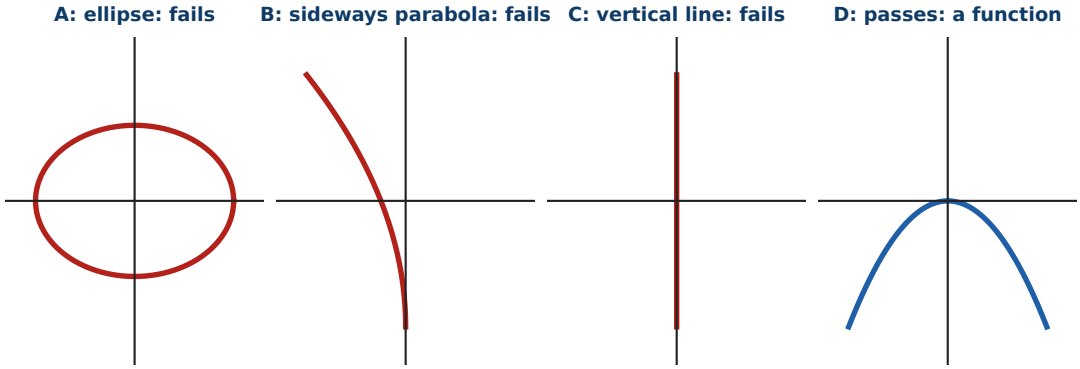
Quick: P25 v **B**; vii **C**. P24 iii **B**; xi **B**. P23 ii **D**; iii **A**.

2 · LINEAR EQUATIONS, SYSTEMS & FUNCTIONS

P25 Q1iv,ix,Q2d,Q3c,Q6a,Q7c · P24 Q1iv,xiv,Q4c,Q6a,Q7b,Q9d · P23 Q1iv,vi,vii,Q2b,Q3b,Q6b

Key ideas

- **Function:** a rule giving **exactly one output for each input**. **Vertical-line test:** a graph is a function if no vertical line meets it twice. In a mapping diagram every input must have **one and only one arrow**.
- **Linear model:** $y = mx + c$: **m = slope =** $\frac{\textbf{rise}}{\textbf{run}}$, c = y-intercept (starting value). Cost models: total = fixed charge + rate × quantity.
- **Slope-intercept form:** solve for y. $5x + 2y = 12 \rightarrow y = -\frac{5}{2}x + 6$. Intercepts: x = 0 gives the y-intercept; y = 0 gives the x-intercept.
- **Systems of equations: substitution** (make one variable the subject) or **elimination** (equalise coefficients and add/subtract). The solution is the **point of intersection** of the two lines.
- **Word problems:** define variables → write equations → solve → check in the problem.



Vertical line test (P24 Q1iv): only D is a function.

Solved questions

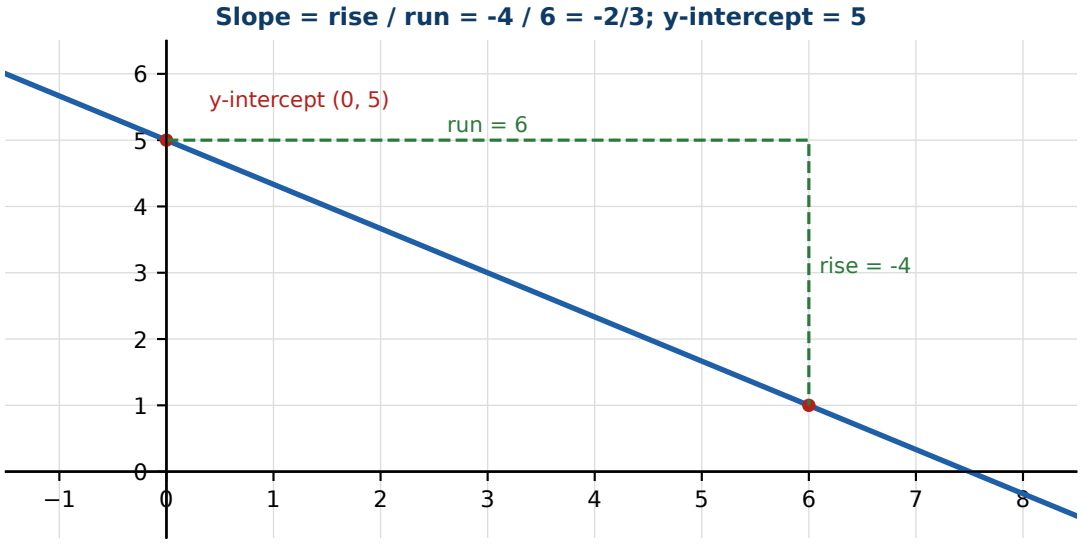
P25 Q1iv [2] She starts with 2 500 and spends 500 each week: **M(w) = 2500 − 500w (A)**.

P25 Q1ix [2] Each week has **exactly one** height, so height is a function of the week: **B**.

P25 Q2d [3] Let the numbers be n and n + 1. Sum = 2n + 1. Difference = 1. Sum = 36 + difference: 2n + 1 = 37 → n = 18. The numbers are **18 and 19**.

P25 Q3c [3] $y = 4x + \frac{1}{3}$ …(1); $3y - 2x = 1$ …(2).

Substitute (1) in (2): $3(4x + \frac{1}{3}) - 2x = 1 \rightarrow 12x + 1 - 2x = 1 \rightarrow 10x = 0 \rightarrow$ **x = 0, y =** $\frac{\textbf{1}}{\textbf{3}}$. Check: $3(\frac{1}{3}) - 0 = 1$ ✓.



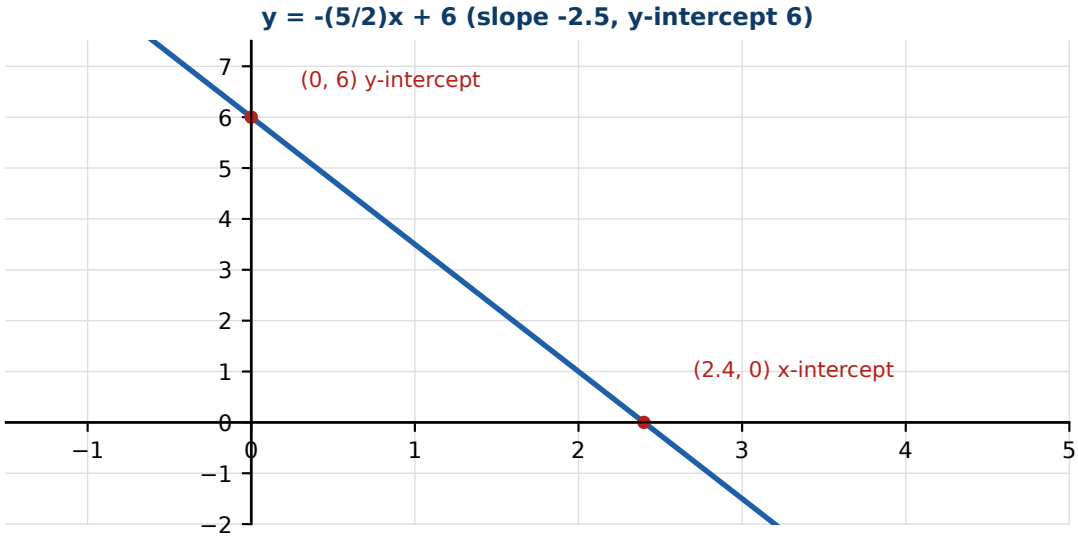
2025 Q6(a): the line passes through (0, 5) and (6, 1).

P25 Q6a [2] From the graph: y-intercept = **5** (the line crosses the y-axis at (0, 5)). Slope = $\frac{\text{rise}}{\text{run}} = \frac{1-5}{6-0} = -\frac{2}{3}$. (Equation $y = -\frac{2}{3}x + 5$.)

P25 Q7c [3] Pema p, son s. Five years ago: $p - 5 = 7(s - 5)$ …(1). In five years: $p + 5 = 3(s + 5)$ …(2). From (1): $p = 7s - 30$. In (2): $7s - 30 + 5 = 3s + 15 \rightarrow 4s = 40 \rightarrow s = 10$, $p = 40$. **Pema is 40 and her son is 10**.

P24 Q1iv / xiv [2+2] iv: the downward parabola passes the vertical-line test → **D**. xiv: $2x + 3y = 5 \rightarrow 3y = 5 - 2x \rightarrow y = \frac{5-2x}{3}$, so **f(x) =** $\frac{\textbf{5 − 2x}}{\textbf{3}}$ **(A)**.

P24 Q4c [3] (i) Total income I = 500 + 150h (h = hours). (ii) h = 8: I = 500 + 150(8) = 500 + 1200 = **Nu 1,700**.



2024 Q6(a)(i): graph of $y = -2.5x + 6$.

P24 Q6a [3] (i) $5x + 2y = 12 \rightarrow 2y = -5x + 12 \rightarrow$ **$y = -\frac{5}{2}x + 6$**: slope -2.5, y-intercept 6. (ii) Sketch: plot (0, 6) and the x-intercept (2.4, 0), join with a ruled line sloping downward (see figure).

P24 Q7b [3] Correct c, incorrect d points deducted. Mipham: $12c - 8d = 40$; Neeshar: $16c - 4d = 60$. Multiply the second by 2: $32c - 8d = 120$. Subtract the first: $20c = 80 \rightarrow c = 4$. Then $48 - 8d = 40 \rightarrow d = 1$. **1 point is deducted for each wrong answer** (4 points for each correct one).

P24 Q9d [2] $2x + 3y = 6$ …(1); $4x - y = 5$ …(2). From (2): $y = 4x - 5$. In (1): $2x + 12x - 15 = 6 \rightarrow 14x = 21 \rightarrow$ **$x = 1.5$, $y = 1$**.

P23 Q1iv [2] h hits and $(20 - h)$ misses: $15h - 5(20 - h) = 60 \rightarrow 20h = 160 \rightarrow h =$ **8 (B)**.

P23 Q1vi / vii [2+2] vi: $f(0) = 5$ and f increases by 5 each time → **f(x) = 5x + 5 (C)**. vii: a mapping is a function if every input has one arrow. In A, input 3 has two arrows; in B, 2 has two; in C, input 2 has none; in D every input has exactly one arrow (1→b, 2→b, 3→c, 4→a) → **D**.

P23 Q2b [3] p pens, q pencils. (i) **$75p + 15q = 600$**. (ii) $p = \frac{600 - 15q}{75} =$ **$8 - \frac{q}{5}$**. (iii) $q = 10$: $p = 8 - 2 =$ **6 pens**.

P23 Q3b [3] Let width w, length $l = 2w + 2$. Perimeter $2(l + w) = 160 \rightarrow l + w = 80 \rightarrow 3w + 2 = 80 \rightarrow w = 26$. **Width 26 m, length 54 m**.

P23 Q6b [3] $\frac{3}{4}x - \frac{2}{3}y = 3$: multiply by 12: $9x - 8y = 36$ …(1). $\frac{1}{2}x - \frac{1}{2}y = 3$: multiply by 2: $x - y = 6 \rightarrow x = y + 6$ …(2). In (1): $9y + 54 - 8y = 36 \rightarrow y = -18$, $x = -12$. Intersection point **$(-12, -18)$**.

Quick: P25 iv **A**; ix **B**. P24 iv **D**; xiv **A**. P23 iv **B**; vi **C**; vii **D**.

3 · QUADRATIC FUNCTIONS & TRANSFORMATIONS

P25 Q1iii,xii,xv,Q5c,d,Q6d,Q9a · P24 Q1x,xii,xviii,Q2c,Q3a,Q8c · P23 Q1v,viii,Q4a,Q7b,Q8b

Three forms of a quadratic

Form	Shows
Standard $f(x) = ax^2 + bx + c$	y-intercept = c; a > 0 opens up, a < 0 opens down
Factored $f(x) = a(x - p)(x - q)$	**roots / x-intercepts p and q**; axis of symmetry $x = \frac{p + q}{2}$
Vertex $f(x) = a(x - h)^2 + v$	vertex (h, v); axis x = h

- **Roots:** factorise or use $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. A quadratic with roots p and q is $(x - p)(x - q) = 0$.
- **Sketching:** find roots, y-intercept (set x = 0), vertex (x halfway between roots), then draw a smooth parabola. Equivalent functions have the same expanded form.

Transformations of $y = x^2$ (mapping notation)

$f(x) = a(x - h)^2 + v$	Mapping (x, y) →
a: vertical stretch/compression; a < 0 reflection in the x-axis	(x, ay)
h: horizontal shift (right if h > 0)	(x + h, y)
v: vertical shift (up if v > 0)	(x, y + v)
all together	(x + h, ay + v)

Solved questions

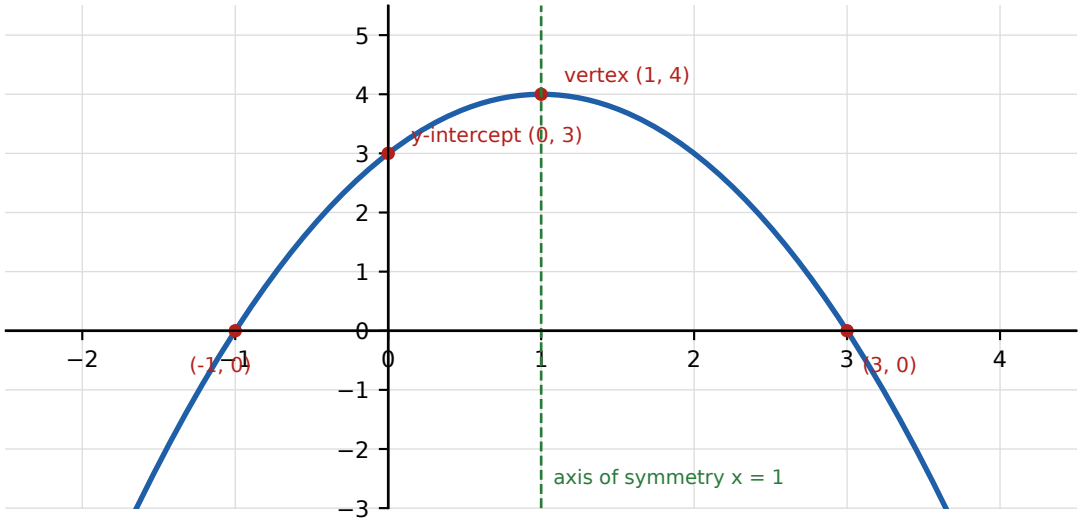
P25 Q1iii [2] C: $x^2 + 6x + 9 = (x + 3)^2$ ✓. Others fail: A: $(x - 2)(x - 3) = x^2 - 5x + 6 \neq x^2 - 5x - 6$; B: $2(x + 1)^2 = 2x^2 + 4x + 2$ but h has no factor 2 and an extra +2; D: $(x + 2)^2 \neq (x - 2)^2$. **C**.

P25 Q1xii [2] Shifted 3 right and 2 up: **$y = (x - 3)^2 + 2$ (A)**.

P25 Q1xv [2] $x^2 - 7x + 10 = (x - 2)(x - 5) = 0 \rightarrow x =$ **2 and 5 (A)**.

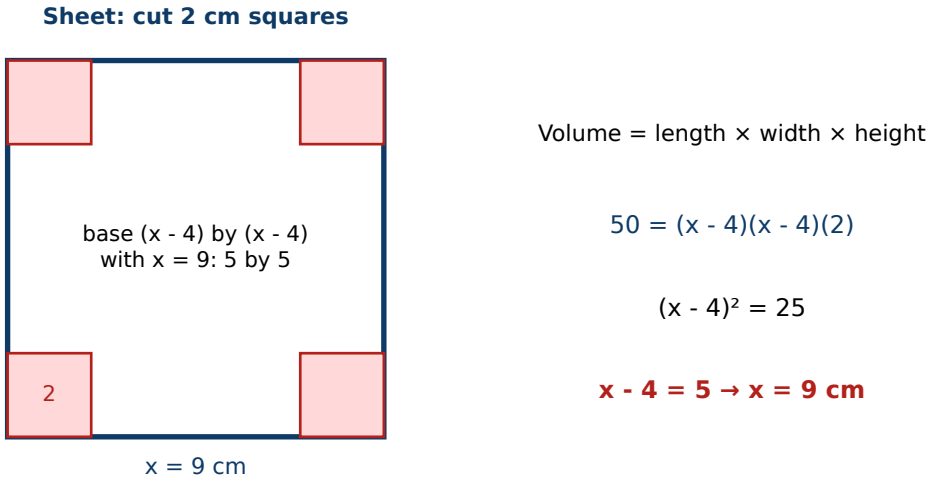
P25 Q5c [2] $h(1) = 64 - 16(1)^2 = 48$ ft > 0, so **no, the object has not reached the ground at t = 1 s**. Ground: $64 - 16t^2 = 0 \rightarrow t^2 = 4 \rightarrow$ **t = 2 seconds**.

f(x) = -(x + 1)(x - 3): opens downward (a = -1)



2025 Q5(d): f(x) = −(x + 1)(x − 3).

P25 Q5d [2] $f(x) = -(x + 1)(x - 3)$: roots x = −1 and 3; axis of symmetry $x = \frac{-1 + 3}{2} = 1$; vertex (1, f(1)) = (1, $-(2)(-2)$) = **(1, 4)**; y-intercept f(0) = **3**; a = −1 so it opens downward. Plot these points and draw a smooth curve (see figure).



2025 Q6(d): the open box.

P25 Q6d [3] Cutting 2 cm squares from a square sheet of side x gives a base (x − 4) by (x − 4) and height 2. V = 2(x − 4)² = 50 → (x − 4)² = 25 → x − 4 = 5 → **x = 9 cm** (x − 4 = −5 is rejected).

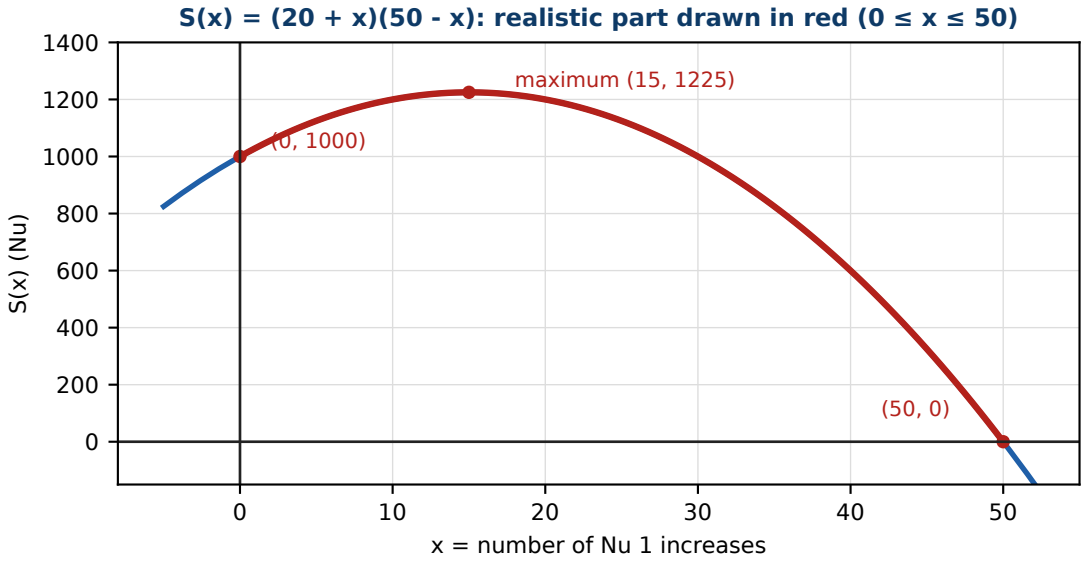
P25 Q9a [2] (x − 4)(x + 3) = x² − x − 12. The equation is **x² − x − 12 = 0**: a = 1, b = −1, c = −12.

P24 Q1x / xii [2+2] x: the curve is a repeating wave → periodic, like the height of a person on a seesaw over time → **C** (a cannon ball gives a parabola). xii: roots −3 and 9: (x + 3)(x − 9) = x² − 6x − 27 = 0 → **A**.

P24 Q1xviii [2] f(x) = 2x² + 4x − 6 = 2(x² + 2x − 3) = 2(x − 1)(x + 3) = g(x) ✓. h(x) = 2(x + 1)² − 3 = 2x² + 4x − 1 ≠ f(x). So **f and g (A)**.

P24 Q2c [3] Width w, length w + 10, area w(w + 10) = 144 → w² + 10w − 144 = 0 → (w + 18)(w − 8) = 0 → w = 8 (reject −18). **Width 8 m, length 18 m**.

P24 Q3a [3] (x, y) → (x + 2, $-\frac{1}{5}$y + 2). (i) **Right 2**; **reflect in the x-axis** and **vertical compression by $\frac{1}{5}$**; then **up 2**. (ii) With f(x) = x²: **g(x) = $-\frac{1}{5}$(x − 2)² + 2**.



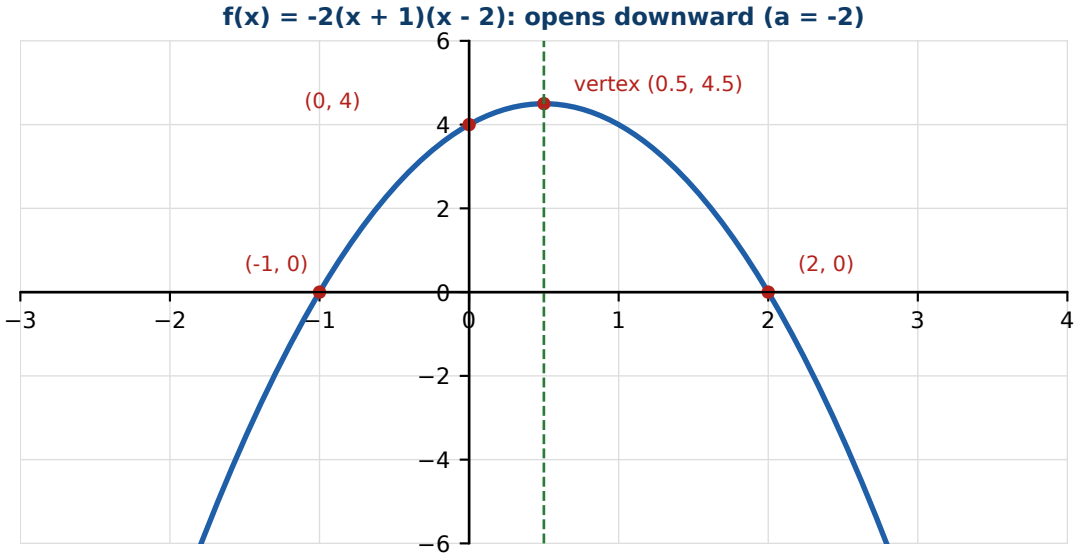
2024 Q8(c): sales function S(x) = (20 + x)(50 − x).

P24 Q8c [4] After x increases of Nu 1: price = 20 + x, number sold = 50 − x. (i) **S(x) = (20 + x)(50 − x) = −x² + 30x + 1000**. (ii) Roots x = −20 and 50; vertex x = 15, S(15) = 35 × 35 = 1225; y-intercept 1000. Sketch for 0 ≤ x ≤ 50: highest sale **Nu 1,225 at a price of Nu 35**.

P23 Q1v / viii [2+2] v: (2x + 3)(4x − 2) = 8(x + $\frac{3}{2}$)(x − $\frac{1}{2}$), so **p = $-\frac{3}{2}$, q = $\frac{1}{2}$ (B)**. viii: x² + (x + 4)² = 20² → x² + 4x − 192 = 0 → (x + 16)(x − 12) = 0 → **12 cm (A)**.

P23 Q4a [4] Width w, length w + 4: w(w + 4) = 60 → w² + 4w − 60 = 0 → (w + 10)(w − 6) = 0 → w = 6. **6 m by 10 m**.

P23 Q7b [3] (i) f(x) = 2x² − 0.5: **(x, y) → (x, 2y − 0.5)**. (ii) −3(x + 1)² − 4: **(x, y) → (x − 1, −3y − 4)**. (iii) (x − 3)² + 6: **(x, y) → (x + 3, y + 6)**.



2023 Q8(b): f(x) = −2(x + 1)(x − 2).

P23 Q8b [3] f(x) = −2(x + 1)(x − 2): roots −1 and 2; vertex at x = 0.5: f(0.5) = −2(1.5)(−1.5) = 4.5; y-intercept f(0) = −2(1)(−2) = 4; opens downward (a = −2). Plot and join with a smooth curve.

Quick: P25 iii **C**; xii **A**; xv **A**. P24 x **C**; xii **A**; xviii **A**. P23 v **B**; viii **A**.

4 · MATRICES & NETWORKS

P25 Q1i,vi,Q4c,Q9c · P24 Q1i,xv,Q3b · P23 Q1i,Q4b,Q9a

Key ideas

- **Order** of a matrix = rows × columns. **Addition / subtraction** need the **same order** P24 Q1i: A; add or subtract matching entries. Scalar multiplication multiplies every entry; the order does not change, so (½A − 2B) for two 3 × p matrices has order **3 × p** P24 Q1xv: C.
- **Identity matrix** I: 1s on the main diagonal, 0 elsewhere. For 2 × 2: [1 0; 0 1] P25 Q1i: B. AI = IA = A.
- **Multiplication** AB needs (columns of A) = (rows of B); the product has order (rows of A) × (columns of B). Each entry = **row × column**. In general **AB ≠ BA**.
- **Applications:** cost = (quantities row) × (price matrix); points = (results table) × (points column).

Networks and adjacency matrices

- **Adjacency matrix:** the entry in row X, column Y = the number of **direct edges (routes) from X to Y**. For an undirected link the entry is 1 in both directions.
- **One stop-over trips** (two edges: X → Z → Y) = entries of the matrix **M² = M × M**.
- To match a matrix with a network, read row by row: row A gives the edges that **leave** A.

Solved questions

P25 Q1vi [2] The matrix has A→B = 2 edges, B→C = 1 edge, C→A = 1 edge. Option D shows two arrows from A to B (the curved arc and the straight line), one arrow B→C and one arrow C→A → **D**. (A, B and C each have a different number or direction of arrows.)

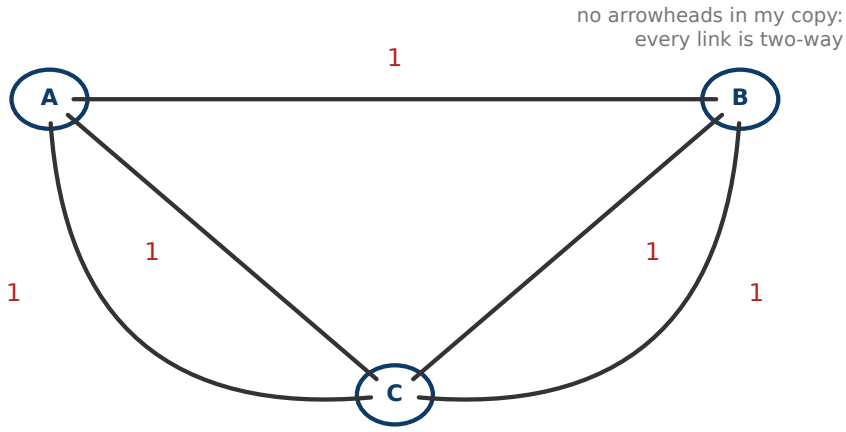
P25 Q4c [2] Quantities [20 40 25] × prices (columns Shop A and Shop B):

[20 40 25] × [950 800; 400 500; 250 350] = [20(950) + 40(400) + 25(250), 20(800) + 40(500) + 25(350)] = [41 250, 44 750]

Shop A costs **Nu 41,250**, Shop B **Nu 44,750**. **Buy from Shop A** (saves Nu 3,500).

P25 Q9c [3] B + C = [0 1; 1 0] + [2 0; 0 2] = [2 1; 1 2]. A(B + C) = [1 2; 3 4][2 1; 1 2] = [1·2 + 2·1, 1·1 + 2·2; 3·2 + 4·1, 3·1 + 4·2] = **[4 5; 10 11]**. (B + C)A = [2 1; 1 2][1 2; 3 4] = [2·1 + 1·3, 2·2 + 1·4; 1·1 + 2·3, 1·2 + 2·4] = **[5 8; 7 10]**. The results are different, so **A(B + C) ≠ (B + C)A**: matrix multiplication is **not commutative**.

A–B: 1 link; A–C: 2 links; B–C: 2 links



2024 Q3(b): the network (links are two-way because my copy shows no arrowheads).

P24 Q3b [3] (i) Order A, B, C. A–B: 1 link; A–C: 2 links (straight line and arc); B–C: 2 links. **M = [0 1 2; 1 0 2; 2 2 0]**. (ii) One stop-over trips = M² = **[5 4 2; 4 5 2; 2 2 8]**. Example: A to B via C = 2 × 2 = 4 trips. **(check)** the figure printed in my copy has no arrows; if your paper shows arrows, fill M with the number of arrows from the row vertex to the column vertex and square it in the same way.

P23 Q1i [2] Count the arrows going from Paro to Phuntsholing: the two curved arrows point to Phuntsholing, and the plain straight line (no arrowhead) is a two-way link, giving **3 services (B)**. **(check)** if the straight line is not counted the answer would be 2.

P23 Q4b [3] (i) Matrix of results (Win, Draw, Loss): **[14 3 1; 10 3 5; 2 1 15]**. (ii) Points per result [3; 1; 0]:

[14 3 1; 10 3 5; 2 1 15] × [3; 1; 0] = [14(3) + 3(1); 10(3) + 3(1); 2(3) + 1(1)] = [45; 33; 7]

Paro FC 45, Druk Lhayul FC 33, Gelephu FC 7 points.

P23 Q9a [2] Add and subtract entry by entry. Top-left: a + (−4) − 2 = −5 → a − 6 = −5 → **a = 1**. Bottom-left: −3 + 2 − 4 = −5 = b → **b = −5**. (Check top-right 2 + 3 − 1 = 4 ✓; bottom-right 4 − 1 − 3 = 0 ✓.)

Quick: P25 i **B**; vi **D**. P24 i **A**; xv **C**. P23 i **B** (check).

5 · TRIGONOMETRY

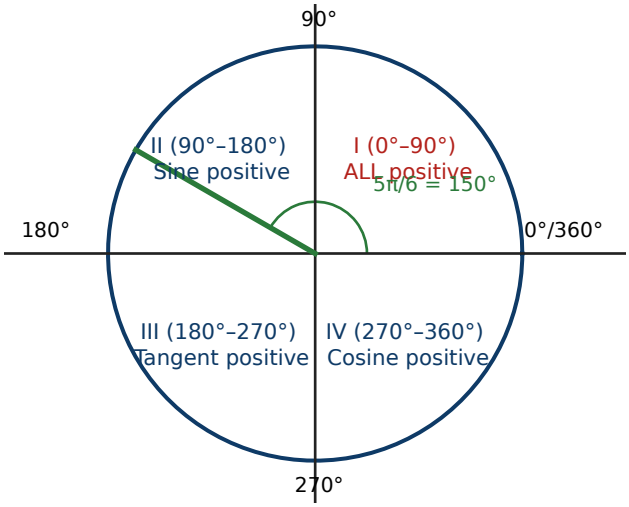
P25 Q1viii,xi,Q2a,Q5b,Q8d · P24 Q1viii,xx,Q5d,Q8a · P23 Q1xii,xiii,Q4c,Q7d

Ratios, identities & special angles

- $\sin\theta = \frac{\text{opp}}{\text{hyp}}$, $\cos\theta = \frac{\text{adj}}{\text{hyp}}$, $\tan\theta = \frac{\text{opp}}{\text{adj}} = \frac{\sin\theta}{\cos\theta}$; $\text{cosec} = \frac{1}{\sin}$, $\sec = \frac{1}{\cos}$, $\cot = \frac{1}{\tan} = \frac{\cos}{\sin}$.
- **Identities:** $\sin^2 + \cos^2 = 1$; $1 + \tan^2 = \sec^2$; **$1 + \cot^2 = \text{cosec}^2$**.
- **Radians:** $180° = \pi$ rad: $30° = \frac{\pi}{6}$, $45° = \frac{\pi}{4}$, $60° = \frac{\pi}{3}$, $90° = \frac{\pi}{2}$, $150° = \frac{5\pi}{6}$ P23 Q1xii: D.

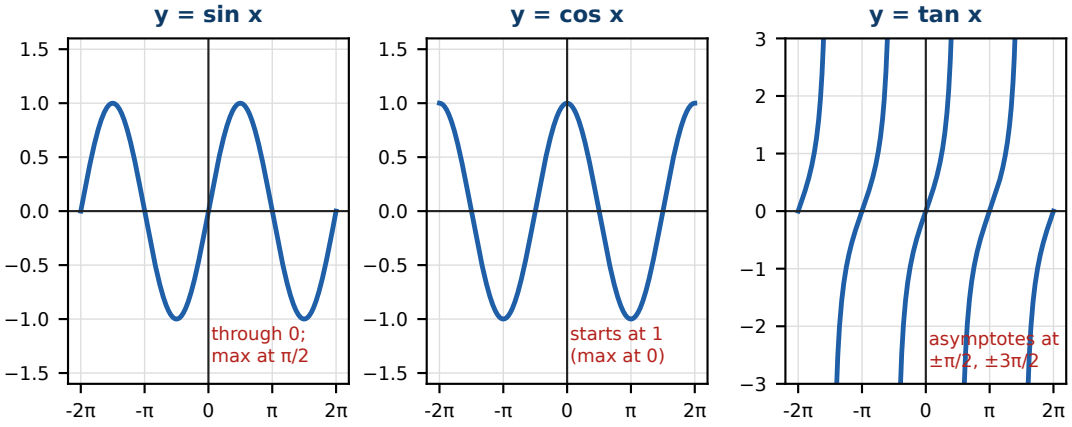
θ	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined
cosec	undefined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
sec	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	undefined
cot	undefined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

Signs of trigonometric ratios (ASTC: All, Sin, Tan, Cos)



Quadrants and the sign of each ratio; the angle 5π/6 = 150° lies in quadrant II.

Graphs of sin, cos, tan



sin x and cos x: waves between −1 and 1 (period 2π); tan x has asymptotes.

Solved questions

P25 Q1viii [2] $\sin\theta = -\frac{1}{\sqrt{2}}$ (negative) and $\tan\theta = 1$ (positive). sin < 0 in quadrants III and IV; tan > 0 in I and III. Both: **third quadrant (B)** (θ = 225°).

P25 Q1xi [2] $\frac{5\pi}{6} = 150°$: from the positive x-axis turn **anticlockwise** through 90° and 60° more, ending in quadrant II close to the negative x-axis. Graph **B** (the arc goes anticlockwise over the top to a shallow ray in quadrant II). A shows the long way clockwise (−240°); C is more than one turn; D is in quadrant IV.

P25 Q2a [2] **$p = \tan 30° = \frac{1}{\sqrt{3}}$ ($= \frac{\sqrt{3}}{3}$); $q = \text{cosec } 60° = \frac{2}{\sqrt{3}}$; $r = \sec 30° = \frac{2}{\sqrt{3}}$; $s = \cot 90° = 0$.** (Use the table above; cosec = 1/sin, sec = 1/cos, cot = cos/sin.)

P25 Q5b [4] (i) $2\cot^2\theta + 2 = 2(\cot^2\theta + 1) = 2\text{cosec}^2\theta = 2 \times \frac{1}{\sin^2\theta} = 2 \times 9 =$ **18**. (ii) $\cos\theta = \frac{5}{4} = 1.25 > 1$. The adjacent side cannot be longer than the hypotenuse, and cos θ always lies between −1 and 1, so **it is not a feasible value**.

P25 Q8d [2] (i) LHS = $\sec x - \tan x \sin x = \frac{1}{\cos x} - \frac{\sin^2 x}{\cos x}$ $= \frac{1 - \sin^2 x}{\cos x} = \frac{\cos^2 x}{\cos x} = \cos x = \frac{1}{\sec x}$ = RHS ✓. (ii) $\tan x \cot x - \sin^2 x = 1 - \sin^2 x = \cos^2 x$ = RHS ✓.

P24 Q1viii [2] $\cot\theta = \frac{4}{3}$ (adj 4, opp 3) → hyp = 5. $\operatorname{cosec}\theta = \frac{\text{hyp}}{\text{opp}} = \frac{5}{3}$ **(D)**.

P24 Q1xx [2] $\sec 30° \tan 60° + \sin 45° \operatorname{cosec} 45° + \cos 30° \cot 60° = \frac{2}{\sqrt{3}}(\sqrt{3}) + 1 + \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{3}} = 2 + 1 + \frac{1}{2} = \frac{7}{2}$ **(D)**.

P24 Q5d [3] (i) $\sec\theta \cot\theta = \frac{1}{\cos\theta} \cdot \frac{\cos\theta}{\sin\theta} = \frac{1}{\sin\theta} = \operatorname{cosec}\theta$ ✓. (ii) Numerator: $1 + \cos\theta - \sin^2\theta = \cos^2\theta + \cos\theta = \cos\theta(\cos\theta + 1)$. So LHS = $\frac{\cos\theta(1 + \cos\theta)}{\sin\theta(1 + \cos\theta)} = \frac{\cos\theta}{\sin\theta} = \cot\theta$ ✓.

P24 Q8a [3] (i) Repeating S-shaped branches with asymptotes: **$y = \tan x$**. (ii) Wave through the origin, rising, maximum 1 at $x = \frac{\pi}{2}$: **$y = \sin x$**. (iii) Wave starting at its maximum 1 when $x = 0$: **$y = \cos x$**.

P23 Q1xiii [2] $\sin\theta = \frac{3}{5} \to \cos\theta = \frac{4}{5}$, $\tan\theta = \frac{3}{4}$. $\tan\theta + \frac{1}{\cos\theta} = \frac{3}{4} + \frac{5}{4} = 2 \to$ **C**.

P23 Q4c [3] $\tan A + \cot A = \frac{\sin A}{\cos A} + \frac{\cos A}{\sin A} = \frac{\sin^2 A + \cos^2 A}{\sin A \cos A} = \frac{1}{\sin A \cos A}$ ✓.

P23 Q7d [3] $\cot A = \frac{13}{5}$: adjacent 13, opposite 5, hypotenuse $= \sqrt{13^2 + 5^2} = \sqrt{194}$. **$\sin A = \frac{5}{\sqrt{194}}$; $\cos A = \frac{13}{\sqrt{194}}$; $\tan A = \frac{5}{13}$; $\operatorname{cosec} A = \frac{\sqrt{194}}{5}$; $\sec A = \frac{\sqrt{194}}{13}$**.

Quick: P25 viii **B**; xi **B**. P24 viii **D**; xx **D**. P23 xii **D**; xiii **C**.

6 · MEASUREMENT: VOLUME, SURFACE AREA & EFFICIENCY

P25 Q1ii,xiv,xvii,xix,Q3b,Q6b,Q7b,d,Q8c · P24 Q1ii,xiii,xvii,xix,Q2b,Q4a,Q6b,Q8b,Q9c · P23 Q1ix,x,xi,xiv,Q2c,Q5b,Q6a,Q8c,Q9b

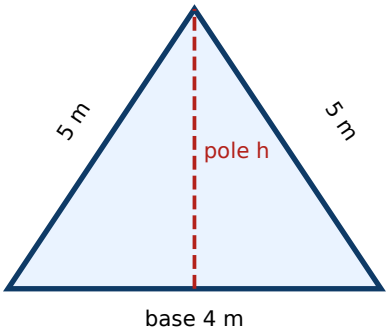
Formulae (given on the formula page)

Solid	Volume	Surface area
Rectangular prism	lwh	$2(lw + wh + lh)$
Cube	e^3	$6e^2$
Any prism	A × h (base area × height)	2A + hP (P = base perimeter)
Cylinder	$\pi r^2 h$	$2\pi r^2 + 2\pi rh$
Pyramid	$\frac{Ah}{3}$	A + area of lateral faces
Cone	$\frac{\pi r^2 h}{3}$	$\pi r^2 + \pi rs$ (s = slant height, $s^2 = r^2 + h^2$)
Sphere	$\frac{4}{3}\pi r^3$	$4\pi r^2$

- **Frustum** (truncated cone) = big cone − small cone, using **similar triangles** to find the missing height. **Composite solids:** add the volumes; for surface area add the exposed faces only (do not count faces joined together).
- **Efficiency:** for the **same volume** the solid with the **smaller surface area** is more efficient (less material); for the **same surface area** the solid with the **larger volume** is more efficient. Cubes and cylinders with h = 2r are the most efficient prisms and cylinders.
- Scaling: if the radius of a cylinder doubles (height same) the volume becomes 2^2 = **4 times**. Displaced volume = base area × rise in water level.
- 1 m^3 = 1 000 000 cm^3; 1 mL = 1 cm^3; 1 L = 1000 cm^3.

Solved questions

2025 Q1(a)(ii): the tent is a triangular prism



tent length 8 m

SA = 2(½×4×h) + (5+5+4)×8
136 = 4h + 112 → h = 6 m

2025 Q1(a)(ii): finding the pole height from the surface area of the tent.

P25 Q1ii [2] Tent = triangular prism with ends of base 4 m and height h, sloping sides 5 m, length 8 m. SA = 2(½ × 4 × h) + (5 + 5 + 4) × 8 = 4h + 112 = 136 → h = **6 m (C)**. **(check)** a true 5-5-4 triangle has height $\sqrt{21} \approx 4.6$ m; the question treats h as unknown, giving 6.

P25 Q1xiv [2] V = $\pi r^2 h$; doubling r gives $\pi(2r)^2 h = 4\pi r^2 h$, so new : original = 4 : 1. The options list 1 : 4, so the intended answer is **C (1 : 4)**. **(check)** printed in the order original : new.

P25 Q1xvii [2] Cube 3 × 3 × 3 = 27 cm^3 plus a hemisphere of radius 1.5 cm: $\frac{2}{3}\pi(1.5)^3 = 7.07$ cm^3. Total = 27 + 7.07 = 34.07 ≈ **34.08 cm^3 (D)**.

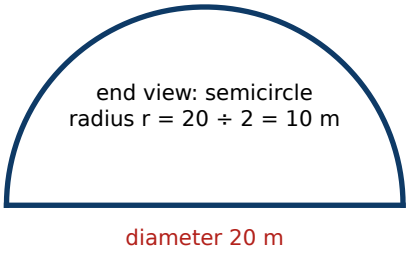
P25 Q1xix [2] Same volume (350 cm^3); surface areas: cube 245, cylinder 160, truncated cone 132 cm^2. A larger surface area means less efficient: cube (245) → cylinder (160) → truncated cone (132). **A**.

P25 Q3b [3] Prism 8 × 10 × 5: SA = 2(8·10 + 10·5 + 8·5) = 2(80 + 50 + 40) = **340 cm^2**. Sphere with the same SA: $4\pi r^2 = 340 \rightarrow r^2 = 27.06 \rightarrow r = 5.20$ cm. **Diameter ≈ 10.4 cm**.

P25 Q6b [3] Cylinder radius = 20 mm ÷ 2 = 1 cm, height 6 cm; cone height = 9 − 6 = 3 cm. V = $\pi(1)^2(6) + \frac{1}{3}\pi(1)^2(3)$ = 6π + π = 7π = **≈ 22 cm^3** (21.99).

P25 Q7b [3] Square pyramid: base side 2h, height h. V = $\frac{1}{3}(2h)^2 h = \frac{4h^3}{3} = 1000 \rightarrow h^3 = 750 \rightarrow h = 9.09$ m. **Height ≈ 9.09 m; base side = 2h ≈ 18.17 m**.

Greenhouse: arch of diameter 20 m, length 80 m



Roof = half of a cylinder's curved surface
= ½ × 2πr × L = π × 10 × 80
= 800π ≈ 2513.3 m²
Cost = 2513.3 × 15 ≈ Nu 37 699

2025 Q7(d): the roof is half the curved surface of a cylinder.

P25 Q7d [2] Roof area = ½ × 2πr × L = π(10)(80) = 800π = 2513.3 m². Cost = 2513.3 × 15 = **≈ Nu 37,699**.

P25 Q8c [2] Sphere radius r; cone base radius 3r, height h. Equal volumes: $\frac{4}{3}\pi r^3 = \frac{1}{3}\pi(3r)^2h = 3\pi r^2h \rightarrow \frac{4}{3}r = 3h \rightarrow$ **$r = \frac{9h}{4} = 2.25h$**.

P24 Q1ii / xiii [2+2] ii: roof slope = $\sqrt{4^2 + 3^2}$ = 5 m; roof = 2 × (5 × 12) = **120 m² (C)**. xiii: 8 cubes of 4 cm: V = 8 × 64 = **512 cm³ (D)**.

P24 Q1xvii [2] Cone r = 3, s = 10. Curved area πrs = π(3)(10) = 94.2 ≈ 94.3 (A); total surface πr(r + s) = 122.5 is not among the options. **A (check)** — the question counts the curved surface (open cone).

P24 Q1xix [2] Compute volume and SA: A (d 5, h 9): V = 176.7, SA = 180.6; B (d 6, h 6.5): V = 183.8, SA = 179.1; C (d 7, h 4.7): V = 180.9, SA = 180.3; D (d 8, h 3.2): V = 160.8, SA = 180.9. Same surface area → the largest volume is most efficient: **B** (also the one with h ≈ 2r).

P24 Q2b [3] Can (r = 3, h = 9): V = π(9)(9) = 254.5 cm³. Tetra pack 5 × 5 × 9: V = 225 cm³. Same price, so Sangay should buy **the can**: it holds **29.5 cm³ more juice** (254.5 > 225). (The can also uses less material: SA 226.2 vs 230 cm².)

P24 Q4a [3] Bottom cube 18 cm, top cube 13 cm; the base is not painted. Bottom: 4 sides 4(18²) = 1296; top ring 18² − 13² = 155. Top cube: 4 sides 4(13²) = 676 and top 13² = 169. Total = 1296 + 155 + 676 + 169 = **2296 cm²**.

P24 Q6b [4] Hexagonal prism: side 3 cm, apothem 2.5 cm, height 9 cm. (i) Lateral area = perimeter × height = (6 × 3) × 9 = 162 cm²; cost = 162 × 0.25 = **Nu 40.50**. (ii) Base area = ½ × perimeter × apothem = ½ × 18 × 2.5 = 22.5 cm²; V = 22.5 × 9 = **202.5 cm³**.

P24 Q8b [3] For a fixed volume, the surface area is smallest for a **cube**: e³ = 729 → e = 9 cm. SA = 6 × 9² = **486 cm²** of wrapper.

P24 Q9c [2] Cuboid 6 × 6 × 5 = 180 cm³; pyramid $\frac{1}{3}$(6 × 6)(6) = 72 cm³. Total = **252 cm³**.

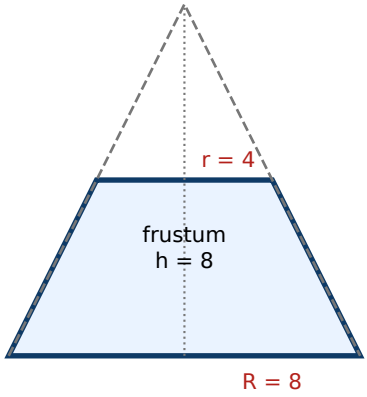
P23 Q1ix / x [2+2] ix: 2(12 × 15 + 15h + 12h) = 684 → 180 + 27h = 342 → h = **6.0 cm (C)**. x: regular pentagon, side 8 cm, apothem 3 cm: base area = 5 × ½ × 8 × 3 = 60 cm²; water rises 5 − 3 = 2 cm: V = 60 × 2 = **120 cm³ (B)** (check the 8 cm and 3 cm labels on your figure).

P23 Q1xi / xiv [2+2] xi: volume 360 cm³, SA = 2s² + $\frac{1440}{s}$: s = 7: 303.7; 6: 312; 5: 338; 4: 392 → smallest area is **7 cm × 7 cm (A)**. xiv: cube volume 343 → edge 7 cm → ball radius = **3.5 cm (B)**.

P23 Q2c [3] Tube r = 5 cm, height 21 cm: one round of paper = 2πrh = 2 × $\frac{22}{7}$ × 5 × 21 = 660 cm². 20 rounds = 13 200 cm². For 8 cubes: 8 × 6e² = 13 200 → e² = 275 → **e ≈ 16.6 cm**. **(check)** depends on reading r = 5 cm and 20 full rounds of the lateral surface.

P23 Q5b [3] Cylinder: diameter 60 cm (r = 30), height 1 m = 100 cm: V = π(30)²(100) = 282 743 cm³. Heating element 12 × 12 × 100 = 14 400 cm³. Water = 282 743 − 14 400 = **268 343 cm³ ≈ 268.3 litres**. **(check)** assumes 60 cm is the diameter and the element is as tall as the container.

Frustum = big cone - small cone (similar triangles)



big cone: R = 8, h = 16
small cone: r = 4, h = 8
V = ⅓π(64)(16) - ⅓π(16)(8)
= (1024π - 128π)/3
≈ 938.3 cm³

2023 Q6(a): the pastry bag (frustum).

P23 Q6a [4] Radii 4 cm (top) and 8 cm (bottom), height 8 cm. Similar triangles: the small cone has height 8, so the big cone has height 16. $V = \frac{1}{3}\pi(8)^2(16) - \frac{1}{3}\pi(4)^2(8) = \frac{1024\pi - 128\pi}{3} =$ **$\approx 938.3\ cm^3$**.

P23 Q8c [3] Walls only (roof not painted): perimeter $2(8 + 12.4) = 40.8$ m, height 5.3 m → area $= 40.8 \times 5.3 = 216.24\ m^2$. Payment $= 216.24 \times 55 =$ **Nu 11,893.20**.

P23 Q9b [4] (i) Pyramid: $V = \frac{1}{3}(10 \times 10)h = 400 \rightarrow h = 12$ cm; slant height 13 cm: SA $= 100 + 4(½ \times 10 \times 13) =$ **360 cm²**. Cuboid $8 \times 4 \times 12.5$: SA $= 2(32 + 100 + 50) =$ **364 cm²**. (ii) Same volume → smaller surface area is more efficient: **the pyramid** (360 < 364), it needs less material.

Quick: P25 ii **C**; xiv **C**; xvii **D**; xix **A**. P24 ii **C**; xiii **D**; xvii **A**; xix **B**. P23 ix **C**; x **B**; xi **A**; xiv **B**.

7 · GEOMETRY: SYMMETRY, TRIANGLE CENTRES & CONSTRUCTIONS

P25 Q1x,xviii,Q3d,Q4b,Q7a,Q8a · P24 Q1v,vii,ix,Q2a,Q4b · P23 Q1xv,xvi,Q2d,Q3c,Q9c

Symmetry

- **Reflection (mirror) symmetry:** a shape folds onto itself along a line of symmetry. **Turn (rotational) symmetry:** the shape fits onto itself when turned less than 360° about a centre; **order of turn symmetry** = number of times it fits in one full turn; angle of turn = 360° ÷ order.
- **Regular polygon with n sides:** n lines of reflection and turn symmetry of order n. **Regular hexagon:** 6 mirror lines (3 through opposite vertices, 3 through midpoints of opposite sides), turn order 6 (60°), and it is also point symmetric (180°) P25 Q7a.
- A figure made of 8 identical leaves around a centre has 8 mirror lines P24 Q1ix: C. A hexagonal nut has turn symmetry of order 6 (A in P23 Q1xv, but the item mentions “three rotations” — (check): if the total number of fits in three turns is meant it is 6 × 3 = 18).
- **Both horizontal and vertical mirror lines:** 4-point star, rhombus, cross, rectangle, bow-tie (not heart, pentagon, trapezium) P25 Q1xviii: B.

Triangle centres

Centre	Made by	Position
Centroid	medians (vertex to midpoint of opposite side)	always inside; balance point
Circumcentre	perpendicular bisectors of the sides	centre of the circle through the 3 vertices; **midpoint of the hypotenuse** in a right triangle; **outside** an obtuse triangle
Orthocentre	altitudes (perpendicular from vertex)	inside acute, at the right-angle vertex in a right triangle, outside obtuse
Incentre	angle bisectors	centre of the inscribed circle

- The point where the three paths from midpoints to the opposite corners meet = **centroid** P24 Q1v: C. Circumcentre of a right-angled triangle = midpoint of the hypotenuse P25 Q1x: C P23 Q1xvi: A.
- **Angle facts:** angles of a triangle add to 180°; an exterior angle = sum of the two opposite interior angles; angles on a line add to 180°.

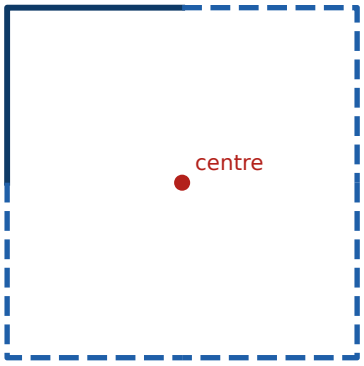
Constructing a triangle and its circumcircle

- Draw the base with a ruler, measure the given angles with a protractor (or construct with compasses), join to form the triangle. Draw the **perpendicular bisectors of two sides** (arcs above and below each side, join); where they meet is the circumcentre O. With radius OA draw the circle through A, B, C. Keep all arcs.

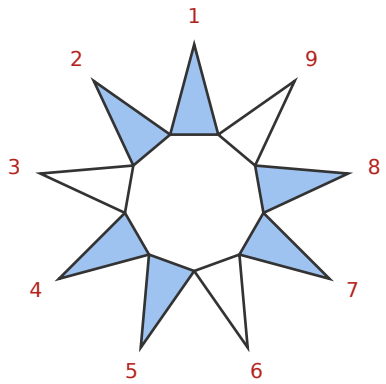
Solved questions

P25 Q3d [3] (i) Order 4 means turn 90° three times about the centre: copy the given corner part into the other three corners (each rotated 90°) to complete the square frame; the right-hand shape is completed by turning it 90° about the centre of its box. (ii) Number the nine points 1–9 and shade **1, 2, 4, 5, 7, 8**: the shaded pattern repeats every 3 points (120°), so the order of turn symmetry is 3.

(i) turn 90° three times
(navy = given part, dashed = added)

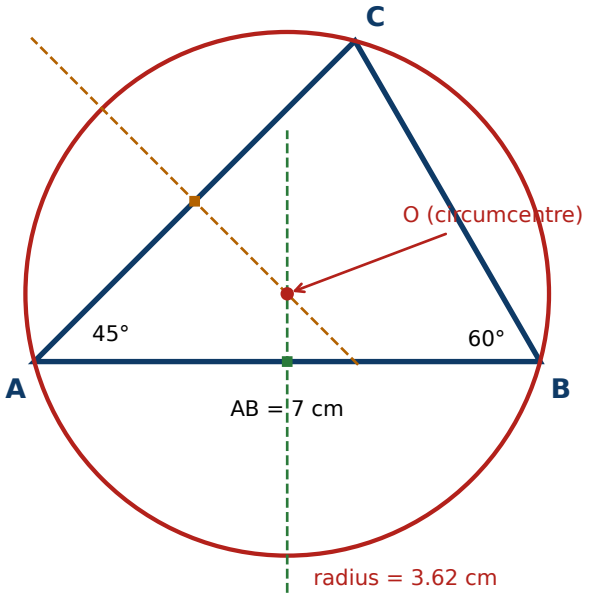


(ii) shade points 1, 2, 4, 5, 7, 8
(pairs repeat every 120°): order 3



2025 Q3(d): completing the pattern.

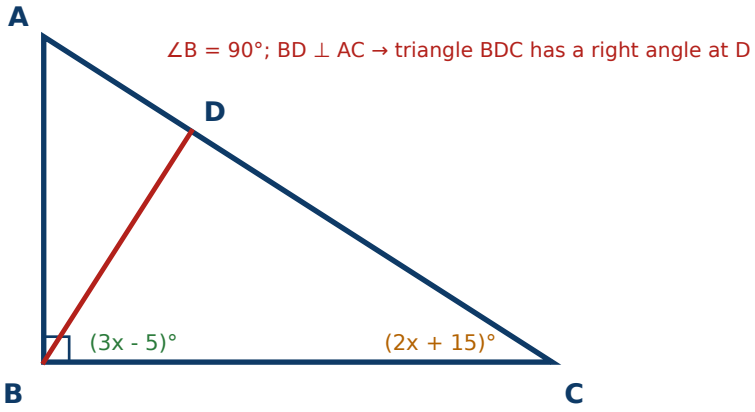
Circumcircle through A, B, C: where the perpendicular bisectors meet



2025 Q4(b): triangle ABC (AB = 7 cm, A = 45°, B = 60°) and its circumcircle.

P25 Q4b [3] Steps: (1) draw AB = 7 cm; (2) at A draw an angle of 45° and at B an angle of 60°; the rays meet at C; (3) draw the perpendicular bisector of AB and of AC (arcs, then ruled lines); (4) they meet at O; (5) with centre O and radius OA draw the circle through A, B and C. Check: C = 75°, radius = $\frac{AB}{2 \sin 75°}$ = **≈ 3.62 cm**.

(3x - 5) + (2x + 15) = 90 → x = 16



2025 Q8(a): BD is an altitude.

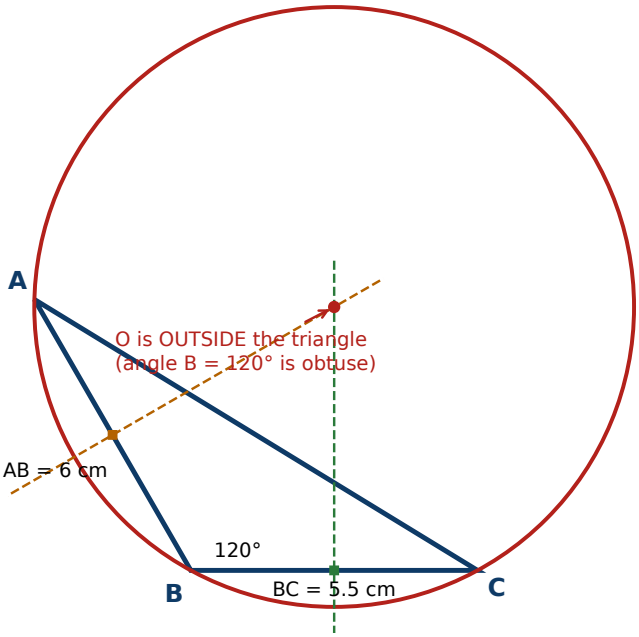
P25 Q8a [2] ∠B = 90° and BD ⊥ AC, so in triangle BDC the angle at D is 90°: ∠DBC + ∠DCB = 90° → (3x − 5) + (2x + 15) = 90 → 5x = 80 → x = 16. **∠DBC = 3(16) − 5 = 43°; ∠DCB = 2(16) + 15 = 47°**.

P25 Q7a [2] A regular hexagon has **6 lines of reflection symmetry** (3 through opposite vertices and 3 through midpoints of opposite sides) and **rotational (turn) symmetry of order 6** (turns of 60° about the centre); it also has point symmetry (180°).

P24 Q1vii [2] Interior angles x, x and y; the exterior angle at the left is 2y. Exterior + adjacent interior = 180°: 2y + x = 180. Triangle: 2x + y = 180. Solve: x = 180 − 2y → 360 − 4y + y = 180 → y = **60° (B)**.

P24 Q2a [4] (i) Order 3: an **equilateral triangle** — centre of rotation = where its medians meet; it fits onto itself after turns of 120°. (ii) Order 4: a **square** — centre = intersection of the diagonals; turns of 90°. Draw the shape and mark the centre with a dot.

Circumcircle of an obtuse triangle (radius 5.75 cm)



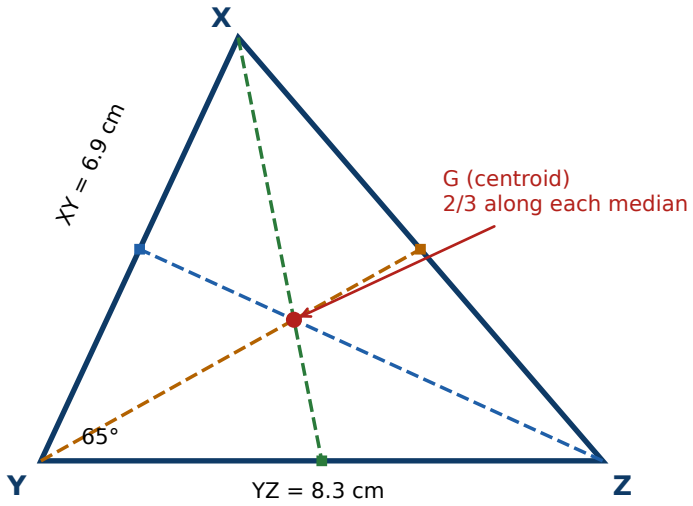
2024 Q4(b): the circumcentre of an obtuse triangle lies outside it.

P24 Q4b [4] Draw BC = 5.5 cm; at B measure 120°; mark A on that ray with BA = 6 cm; join AC (≈ 9.96 cm). Draw the perpendicular bisectors of BC and BA; they meet at O, which lies **outside** the triangle because angle B is obtuse. Circle with centre O through A, B and C: **radius ≈ 5.75 cm**.

P23 Q1xv / xvi [2+2] xv: a hexagonal nut fits onto itself 6 times in each full turn, so **A (6) (check)**. xvi: the circumcentre of a right triangle is at the midpoint of the hypotenuse → **A** (the centroid is the intersection of medians; the orthocentre is of altitudes).

P23 Q2d [2] The twisted prism fits onto itself after a half turn (180°) about its vertical axis: **turn symmetry of order 2** about the vertical line through its centre. **(check)** read from the figure.

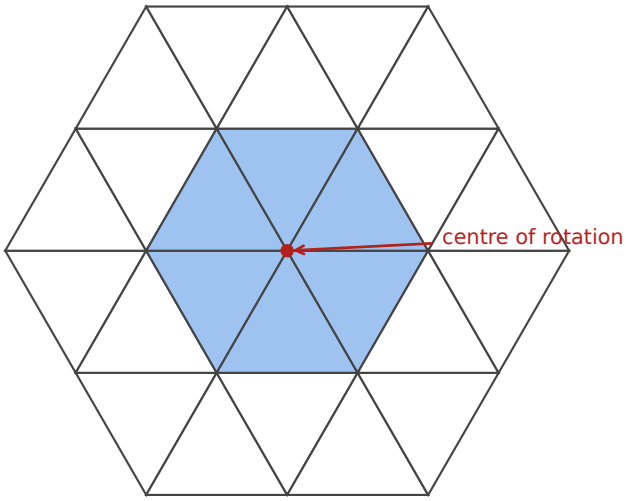
Medians join each vertex to the midpoint of the opposite side



2023 Q3(c): triangle XYZ and its centroid G.

P23 Q3c [4] Draw YZ = 8.3 cm; at Y measure 65°; mark X on that ray with YX = 6.9 cm; join XZ. Find the midpoints of two sides by constructing their perpendicular bisectors (or measuring); join each midpoint to the opposite vertex (two medians). Their intersection is the **centroid G** (it lies 2/3 of the way from each vertex).

Six triangles round the centre: order of turn symmetry 6 (turn 60°)



2023 Q9(c): shading six triangles for turn symmetry of order 6.

P23 Q9c [2] The hexagon is split into 24 triangles around its centre. Shade the **six triangles that meet at the centre** (or one triangle in the same position in each of the six 60° sectors). The pattern fits onto itself after every 60° turn: **order of turn symmetry 6**.

Quick: P25 x **C**; xviii **B**. P24 v **C**; vii **B**; ix **C**. P23 xv **A** (check); xvi **A**.

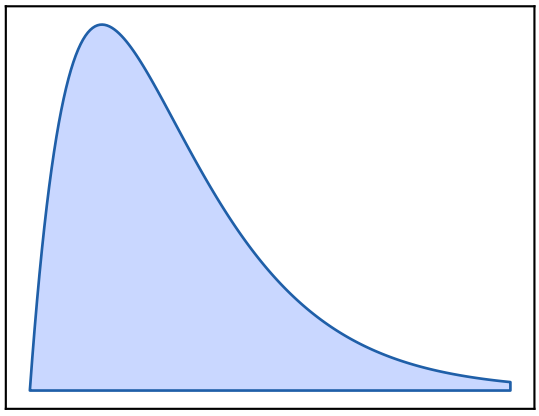
8 · DATA MANAGEMENT & PROBABILITY

P25 Q1xiii,xvi,xx,Q2c,Q3a,Q4a,Q5a,Q8b,Q9b · P24 Q1vi,xvi,Q3c,Q5a,b,Q7a,Q9b · P23 Q1xvii–xx,Q3a,Q5c,Q6c,Q7c,Q8d,Q9d

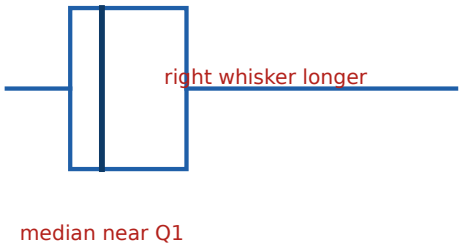
Statistics: key ideas

- **Five-number summary:** minimum, Q_1, median (Q_2), Q_3, maximum. For ungrouped data: sort, find the median; Q_1 = median of the lower half, Q_3 = median of the upper half (leave out the median when n is odd). **Range** = max − min; **IQR** = $Q_3 - Q_1$.
- **Grouped data:** Median = $L + \frac{i}{f}(\frac{n}{2} - c)$ — L lower boundary of the median class, i class width, f its frequency, c cumulative frequency before it. Q_1 uses $\frac{n}{4}$, Q_3 uses $\frac{3n}{4}$. Mean = $\frac{\Sigma fx}{\Sigma f}$.
- **Box plot:** box from Q_1 to Q_3, line at the median, whiskers to min and max. Equal whiskers and a central median → symmetric. **Right-skewed** (long tail to the right): median closer to Q_1 and the right whisker longer P25 Q1xx: D. A **smaller spread** (range, IQR) means more consistent / reliable data.
- **Outliers** change the **mean** a lot but the **median** hardly at all.
- **Scatter plot & correlation:** points rising → positive; falling → negative; no pattern → none; the closer to a line the stronger (r near ±1). The **line of best fit** passes through the middle of the points (about equal points above and below) and is used for predictions. Independent variable on the x-axis.

Right-skewed histogram



Box plot of the same data



Skewed histogram and its box plot.

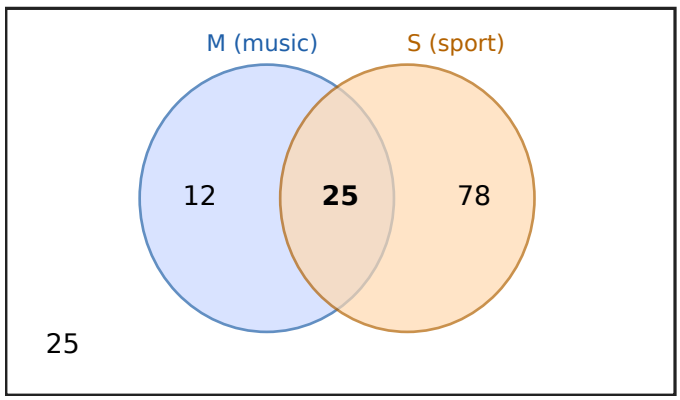
Probability: key ideas

- P(event) = $\frac{\text{favourable outcomes}}{\text{total outcomes}}$. **Independent events:** one does not change the other (with replacement; separate rolls): P(A and B) = P(A) × P(B). **Dependent events** (without replacement): P(A and B) = P(A) × P(B | A). **Conditional:** P(B | A) = $\frac{P(A \text{ and } B)}{P(A)}$.
- A dependent event **changes the sample space** and so changes the probability of the next event P25 Q3a.
- **Venn diagram:** for P(M given S) use only the S circle as the new total.

Solved questions — Section A

P25 Q1xiii / xx [2+2] xiii: the points rise and are fairly close to a line → **close to 1 (B)** (not exactly 1). xx: right-skewed → median near the lower quartile, right whisker longer → **D**.

Given S, the sample space is only the S circle (103 students)



P(M | S) = 25 / (25 + 78) = 25/103

2025 Q1(xvi): conditional probability from a Venn diagram.

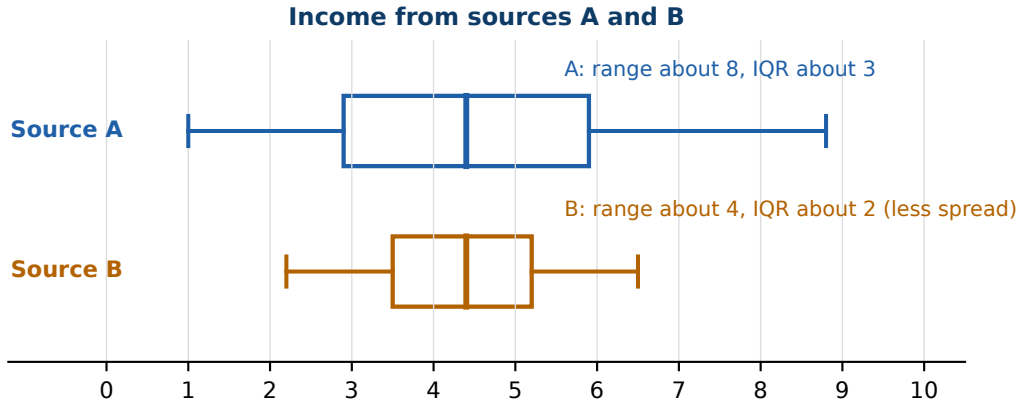
P25 Q1xvi [2] Given that the student plays a sport, the total is 25 + 78 = 103; of these 25 study music. P(M | S) = $\frac{25}{103}$ **(C)**.

P24 Q1vi / xvi [2+2] vi: total = 8 + 2 + 4 + 6 = 20; archery = 8 + 2 = 10 → P = $\frac{10}{20}$ **(D)**. xvi: the points rise steeply along a line → **close to 1 (D)**.

P23 Q1xvii [2] n = 200, $\frac{n}{2}$ = 100. Cumulative: 28, 74, 128 → median class 2000–3000 (c = 74, f = 54, L = 2000, i = 1000). Median = 2000 + $\frac{1000}{54}(100 - 74)$ = **Nu 2481.48 (B)**.

P23 Q1xviii–xx [2+2+2] xviii: as time passes the distance left **decreases** → **D**. xix: P(both) = 0.2 × 0.6 = **0.12 (A)**. xx: independent if the second roll does not depend on the first: "3 or 4 on the first roll" and "number less than 5 on the second roll" → **C**.

Solved questions — Section B



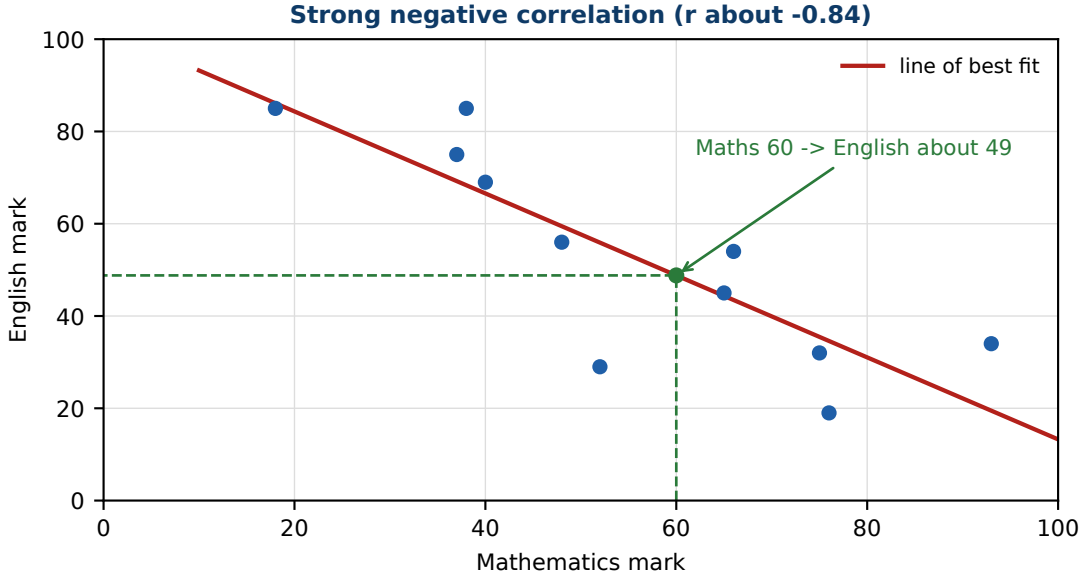
2025 Q2(c): values read from the box plots.

P25 Q2c [2] Both have about the same median (≈ 4.4), but source A has a much larger spread (range ≈ 8, IQR ≈ 3) than B (range ≈ 4, IQR ≈ 2). **Source B is more reliable** because its values are more consistent (less spread).

P25 Q3a [1] When an event is **dependent**, the first outcome **changes the number of possible outcomes**, so the probability of the second event **changes** (P(B | A) ≠ P(B)), e.g. drawing cards without replacement.

P25 Q4a [2] MEASUREMENTS has 12 letters with E three times. (i) With replacement: P(E) = $\frac{3}{12} = \frac{1}{4}$. (ii) Without replacement: $\frac{2}{11}$. (iii) **The events in question (ii)** are dependent (the second probability changes after the first draw).

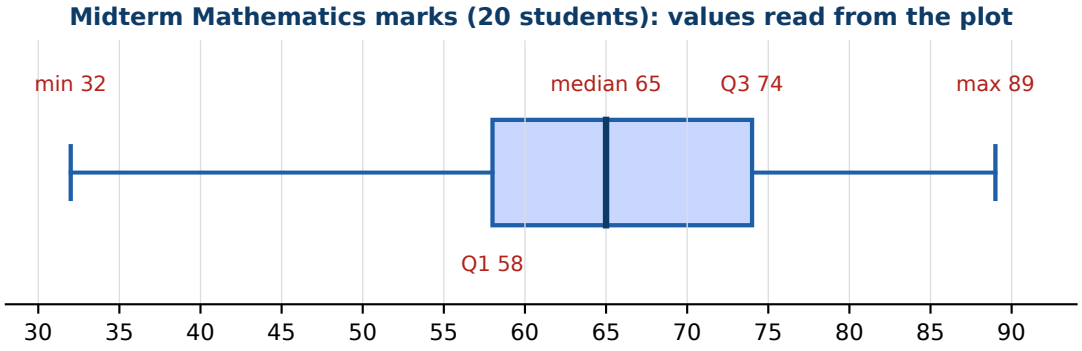
P25 Q5a [2] The outlier 98 pulls the **mean upward** (increases it), but the **median hardly changes** because it depends only on the middle values: the median is resistant to outliers.



2025 Q8(b): scatter plot of Maths and English marks with the line of best fit.

P25 Q8b [4] (i) Plot the 11 points (Maths on the x-axis, English on the y-axis). (ii) Draw a ruled line through the middle of the points (about y = −0.89x + 102); at Maths = 60 read English ≈ **49 marks** (accept 45–52). (iii) As Maths goes up English goes down: **strong negative correlation** (r ≈ −0.84).

P25 Q9b [3] Sorted: 5, 8, 10, 15, 20, 25, 30 (n = 7). Median = 15. Upper half 20, 25, 30 → **Q_3 = 25**. It means **75% of the students scored 25 or less** (only the top 25% scored above 25).



2024 Q7(a): box plot of the midterm marks.

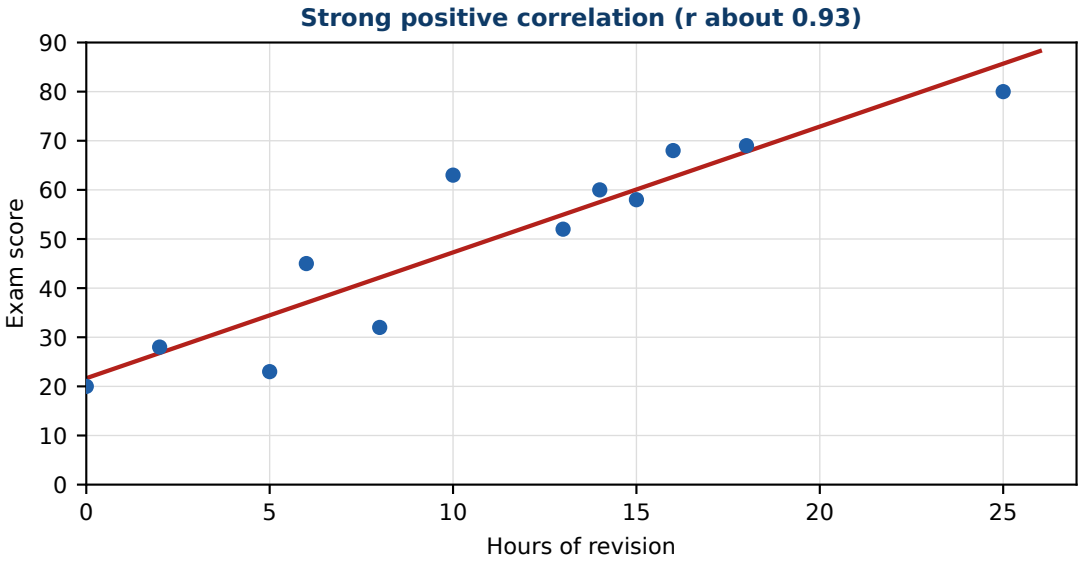
P24 Q7a [4] (i) The median (65) is close to the middle of the box and the left whisker (26) is longer than the right whisker (15): **approximately symmetric, slightly skewed to the left (negatively skewed)**. (ii) **Median = 65 marks**. (iii) Any two: lowest mark about 32 and highest about 89; range ≈ 57; the middle 50% scored between 58 and 74 (IQR 16); 25% scored above 74; the lower marks are more spread out than the higher marks. **(check)** values read from the plot.

P24 Q3c [4] Back-to-back (double) stem and leaf plot, stems in tens (leaves = units):

Messi (leaves)	Stem	Ronaldo (leaves)
	2	3 6
	3	3
5 1 1	4	2 2 2
8 4 3 1	5	1 5
1 0	6	0 1
3	7	

(ii) Modes: **Messi 41; Ronaldo 42**. (iii) Ranges: Messi 73 − 41 = **32**; Ronaldo 61 − 23 = **38**.

P24 Q5a,b [4] POSSIBLE has 8 tiles, 3 vowels (O, I, E). (i) Replaced: $\frac{3}{8}$. (ii) Not replaced: $\frac{2}{7}$. (b) P(Maths | English) = $\frac{\text{P(both)}}{\text{P(English)}} = \frac{0.6}{0.8} = \mathbf{0.75}$.



2024 Q9(b): hours of revision against exam score.

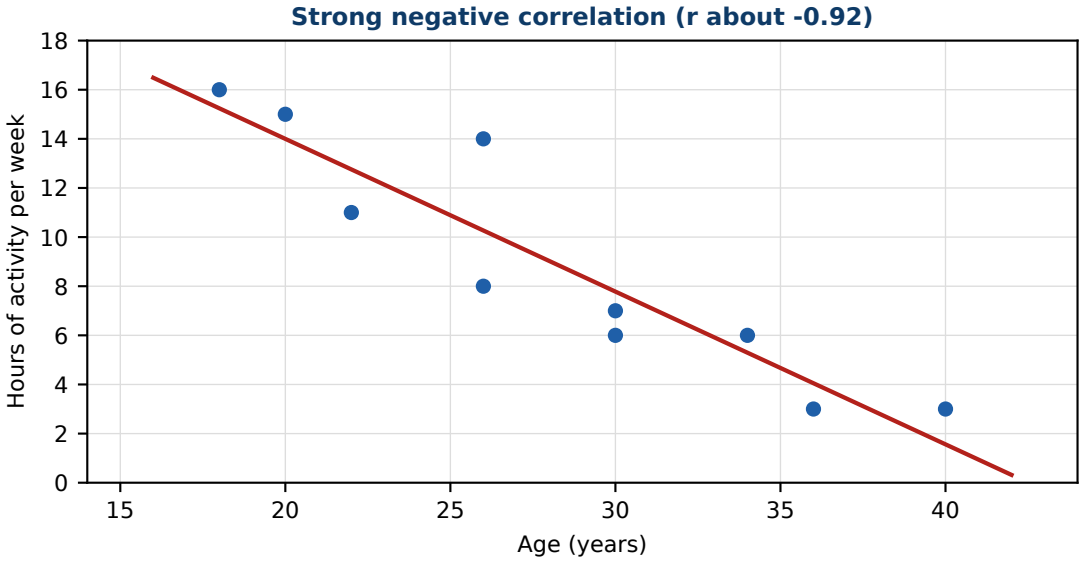
P24 Q9b [3] (i) Plot the 12 points (hours on the x-axis, score on the y-axis). (ii) The scores rise as the hours of revision increase: **strong positive correlation** (r ≈ 0.93); students who revised longer generally scored higher.

P23 Q3a [3] (i) Heights in cm: stems 14–18, leaves = units.

Girls	Stem	Boys
5	14	
7 5 5 4	15	
8 5 4 2 0	16	7 5 5 2 0
1 0	17	9 7 6 2 2 0
	18	0

(ii) **The boys are generally taller than the girls**: median 171 cm against 161 cm (means 170.4 and 160.5), and boys' heights are concentrated in 160–180 cm while the girls' are in 145–171 cm.

P23 Q5c [3] Total = 17 + 12 − 5 + 2 = 26. Only cricket = 17 − 5 = 12; only football = 12 − 5 = 7; both = 5. (i) $\frac{12}{26} = \frac{6}{13}$. (ii) $\frac{7}{26}$. (iii) $\frac{5}{26}$.



2023 Q6(c): age and weekly activity.

P23 Q6c [3] (i) Plot the 10 points (age on the x-axis). (ii) Older people do fewer hours of activity: **strong negative correlation** ($r \approx -0.92$).

P23 Q7c [2] Sorted: 9, 11, 11, 14, 17, 18, 19, 20, 20, 22. Min = 9; Q_1 = median of 9, 11, 11, 14, 17 = 11; median = $\frac{17 + 18}{2} = 17.5$; Q_3 = median of 18, 19, 20, 20, 22 = 20; max = 22. **Five-number summary: 9, 11, 17.5, 20, 22**.

P23 Q8d [2] (i) **Independent variable: wind speed; dependent variable: mechanical power**. (ii) The points curve upward: power rises faster and faster as the wind speed increases → **a non-linear (curved, increasing) relationship**, positive correlation.

P23 Q9d [2] From the bag (counted from the figure): 6 black and 9 white counters, 15 in all. P(black then white, no replacement) = $\frac{6}{15} \times \frac{9}{14} = \frac{54}{210} = \frac{9}{35}$. **(check)** recount the counters on your paper and use the same method.

Quick: P25 xiii **B**; xvi **C**; xx **D**. P24 vi **D**; xvi **D**. P23 xvii **B**; xviii **D**; xix **A**; xx **C**.

SECTION A ANSWER KEY — 2025, 2024 & 2023

20 MCQs each, 2 marks each · (check) = printing fault or reading of a figure

2025 — Q1 [20 × 2]

No.	Ans	Why (short)
i	**B**	identity matrix [1 0; 0 1]
ii	**C**	136 = 4h + 112 → h = 6 m
iii	**C**	$x^2 + 6x + 9 = (x + 3)^2$
iv	**A**	M(w) = 2500 − 500w
v	**B**	l + w = 6√3, l = 4√3 → w = 2√3
vi	**D**	A→B twice, B→C, C→A
vii	**C**	3500 / 10 000 = 35%
viii	**B**	sin < 0, tan > 0 → quadrant III
ix	**B**	each week has one height
x	**C**	circumcentre = midpoint of hypotenuse
xi	**B**	5π/6 = 150°, anticlockwise to quadrant II
xii	**A**	$y = (x - 3)^2 + 2$
xiii	**B**	rising, close to 1
xiv	**C** (check)	radius volume × 4 (option 1 : 4 is old : new)times; 2 gives volume volume × 4 (option 1 : 4 is old : new)times; 4
xv	**A**	(x − 2)(x − 5) = 0
xvi	**C**	25 / (25 + 78) = 25/103
xvii	**D**	27 + hemisphere 7.07 ≈ 34.08
xviii	**B**	star, rhombus, cross have both mirror lines
xix	**A**	larger SA = less efficient: cube, cylinder, cone
xx	**D**	right skew: median near Q1, right whisker long

2024 — Q1 [20 × 2]

No.	Ans	Why (short)
i	**A**	same order to add
ii	**C**	2 × (5 × 12) = 120 m²
iii	**B**	$(1 + r/2)^6 = 1.2653$ → r = 8%
iv	**D**	vertical line test
v	**C**	centroid
vi	**D**	(8 + 2) / 20
vii	**B**	2y + x = 180, 2x + y = 180 → y = 60°
viii	**D**	cosec = 5/3
ix	**C** (check)	8 leaves → 8 mirror lines
x	**C**	periodic wave: seesaw
xi	**B**	21√(2y)
xii	**A**	$(x + 3)(x - 9) = x^2 - 6x - 27$
xiii	**D**	8 × 64 = 512
xiv	**A**	y = (5 − 2x)/3
xv	**C**	order 3 × p
xvi	**D**	strong positive
xvii	**A** (check)	πrs = 94.3 (curved surface)
xviii	**A**	f and g are equal
xix	**B**	same SA: largest volume
xx	**D**	2 + 1 + 1/2 = 7/2

2023 — Q1 [20 × 2]

No.	Ans	Why (short)
i	**B** (check)	2 arrows + 1 two-way link = 3
ii	**D**	$\sqrt[3]{40} = 2\sqrt[3]{5}$
iii	**A**	6% simple gives the most (Nu 300)
iv	**B**	15h − 5(20 − h) = 60 → h = 8
v	**B**	p = −3/2, q = 1/2
vi	**C**	f(x) = 5x + 5
vii	**D**	one arrow from each input
viii	**A**	$x^2 + (x + 4)^2 = 400$ → 12 cm
ix	**C**	684 = 2(180 + 27h) → h = 6
x	**B** (check)	base 60 cm² × 2 cm rise
xi	**A**	smallest SA for the same volume
xii	**D**	90° = π/2
xiii	**C**	3/4 + 5/4 = 2
xiv	**B**	edge 7, radius 3.5
xv	**A** (check)	hexagon: order 6
xvi	**A**	circumcentre at midpoint of hypotenuse
xvii	**B**	2000 + (1000/54)(26) = 2481.48
xviii	**D**	distance left falls with time
xix	**A**	0.2 × 0.6 = 0.12
xx	**C**	second roll independent of the first

LAST-DAY CHECKLIST

tick when you can do it without looking

- ☐ **Surds & finance:** simplify and combine radicals; entire radicals; compound interest $A = P(1 + r/n)^{nt}$ (find r by taking the root); dividend DA = fv × r × n; yield = DA / OI; premium; compare rates by effective annual rate.
- ☐ **Linear:** write models (M = 2500 − 500w); slope and y-intercept from a graph or equation; solve two equations (substitution / elimination); age, perimeter and points problems; function = one output per input.
- ☐ **Quadratics:** factored form roots; expand and compare for equivalent functions; vertex, y-intercept, axis; sketch; roots to equation; transformations and mapping notation (x + h, ay + v); word problems (area, box, sales, height of an object).
- ☐ **Matrices:** same order to add; multiplication rule and order; cost / points tables; A(B + C) vs (B + C)A; adjacency matrix and M^2 for one stop-over trips.
- ☐ **Trig:** special-angle table; ratios from a given ratio; quadrants and signs; radians; identities and proofs ($1 + \cot^2 = \text{cosec}^2$); graphs of sin, cos, tan.
- ☐ **Solids:** cylinder, cone, pyramid, sphere, prism; composite and frustum; displaced volume; efficiency (smaller SA for the same volume); scale factor (radius × 2 → volume × 4).
- ☐ **Geometry:** mirror lines and turn symmetry of regular polygons; centroid, circumcentre (right and obtuse triangles); construct a triangle and its circumcircle; triangle and exterior angles.
- ☐ **Data:** five-number summary and quartiles; grouped-data median; box plots and skew; double stem-and-leaf; scatter plot and line of best fit; correlation words; effect of an outlier.
- ☐ **Probability:** with and without replacement; conditional from a Venn diagram; two-event multiplication; dependent vs independent.
- ☐ **In answers:** formula, substitution, answer with unit; draw graphs with scale and label; keep construction arcs; give a reason for “justify”.